(a) Scenario file scenarios/drift_example.yaml

```yaml
name: thermostat_drift
start: 2023-07-10
end: 2023-07-16
realism: 1.0
occupancy: {stochastic: true}
injections:
  - type: sensor_drift
    target: ZN-Core_mid/ZN-T
    start: 2023-07-11T00:00:00
    params: {rate_per_day: 0.5, max_abs: 4.0}
```

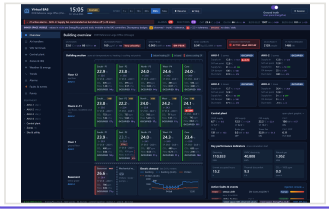
(b) REST API virtualbas/server/app.py

```
B=http://127.0.0.1:8000
curl --json '{"type": "sensor_bias",
  "target": "AHU-1/SA-T", "params": {"bias": 2},
  "duration": "1d", "source": "ai"}' \
  $B/api/injections
# observed value and EnergyPlus truth
curl "$B/api/read?points=AHU-1/SA-T&truth=true"
# operator override at priority 8
curl --json '{"point": "AHU-1/SA-T-SP",
  "value": 14, "priority": 8}' $B/api/write
```

loaded as a session

HTTP and WebSocket

(e) Web GUI



Virtual BAS

one running building:
point list, historian,
alarms, fault catalog,
clock 1× … max

Priority array lowest number wins



MCP server calls REST

in-process, lock-step

(c) LLM agent via MCP virtualbas/ai/mcp_server.py

```
# agent decision of Experiment 5 as tool calls
read_points(points=["ZN-Core_bottom/ZN-T"])
inject(type="sensor_drift",
       target="ZN-Core_bottom/ZN-T",
       params={"rate_per_day": 0.8,
               "max_abs": 3},
       duration="2d",
       note="slow stealthy drift "
            "of a zone thermostat")
# -> {"id": "inj-0001", "source": "ai", ...}
```

(d) Gymnasium examples/04_gym_agent.py

```
env = VirtualBASEnv("buildings/large_office",
        scenario=sc, decision_steps=3)
obs, info = env.reset(seed=0)
while True:
    oat = obs[env.obs_points.index("WX/OA-T")]
    sat = np.clip(16.5 - (oat - 15) * 0.4,
                  12.8, 16.5)
    obs, reward, terminated, truncated, info = \
        env.step(np.array([sat] * 4 + [6.7]))
    if terminated: break
```